Lasserre hierarchy

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When developing global solvers for applications in Robotics, we typically reformulate or relax a least squares problem to a non-convex quadratically constrained quadratic program (QCQP). We then apply Shor's relaxation to get a convex semidefinite program (SDP) and solve it by off-the-shelf SDP solvers such as MOSEK or low-rank SDP methods such as the Burer-Monteiro method (Riemannian staircase). However, rather than reformulating/relaxing the problem to a QCQP, we consider reformulating it in to a polynomial optimization problem (POP), then apply the moment relaxation to obtain a convex SDP. This allows us to deal with a broader class of problems without restricting ourselves with QCQPs. Specifically, POP with degree ≤ 2 is equivalent to some QCQP since we can write polynomials of degree 2 as a quadratic function:

$$ax_1^2 + 2bx_1x_2 + cx_2^2 + dx_1 + ex_2 + f \iff x^\top \begin{pmatrix} a & b \\ b & c \end{pmatrix} x + \begin{pmatrix} d \\ e \end{pmatrix}^\top x + f.$$

In this note, we first review the Shor's relaxation for QCQPs, then the Lasserre's hierarchy (moment relaxation) for general POPs.

1 Shor's relaxation

1.1 Non-convex QCQP to convex SDP

Consider a non-convex quadratically constrained quadratic program (QCQP):

$$\begin{aligned} \min_{x\in\mathbb{R}^n} \ & x^\top C x \\ \text{s.t. } & x^\top A_i x = b_i, \ i = 1, \dots, m \end{aligned} \tag{1.1}$$

Since $x^\top M x = Mxx^\top = \langle M, xx^\top \rangle$, we lift the vector variable into a matrix $X := xx^\top$, where X is positive semidefinite and rank 1, i.e., $X \succeq 0$ and $\text{rank}(X) = 1$. Therefore, Problem (1.1) is equivalent to a non-convex SDP:

$$\begin{aligned} \min_{X\in\mathbb{R}^{n\times n}} \ & \langle C, X \rangle \\ \text{s.t. } & \langle A_i, X \rangle = b_i, \ i = 1, \dots, m \\ & X \succeq 0 \\ & \text{rank}(X) = 1 \end{aligned}$$

where the non-convexity comes from the rank constraint. If we relax the rank constraint, we obtain the following convex SDP:

$$\begin{aligned} \min_{X\in\mathbb{R}^{n\times n}} \ & \langle C, X \rangle \\ \text{s.t. } & \langle A_i, X \rangle = b_i, \ i = 1, \dots, m \\ & X \succeq 0 \end{aligned} \tag{1.2}$$

We immediately have a sufficient condition for the tightness of relaxation:

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Theorem 1.1

Let X^* be a global minimum of Problem (1.2). If X^* is rank 1, then the relaxation is tight and x^* is a global minimum of Problem (1.1), where $X^* = (x^*)(x^*)^\top$.

In practice, we solve Problem (1.2) and check whether the solution is rank 1 or not, then exactly recover or approximate the solution of Problem (1.1). Note that we only consider quadratic functions without linear and constant terms. However, one can reformulate any quadratic function to a linear matrix function as follows:

$$x^\top A x + 2b^\top x + c \iff \left\langle \begin{pmatrix} A & b \\ b^\top & c \end{pmatrix}, \begin{pmatrix} xx^\top & x \\ x^\top & 1 \end{pmatrix} \right\rangle.$$

1.2 Dual problems

The Lagrangian and dual function of Problem (1.1) are

$$\begin{aligned} \mathcal{L}(x, \lambda) &= x^\top C x - \sum_{i=1}^m \lambda_i (x^\top A_i x - b_i), \\ g(y) &= \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \lambda) = \sum_{i=1}^m \lambda_i b_i + \inf_{x \in \mathbb{R}^n} x^\top \underbrace{\left(C - \sum_{i=1}^m \lambda_i A_i \right)}_{S} x = \begin{cases} b^\top \lambda & \text{if } S \succeq 0, \\ -\infty & \text{otherwise.} \end{cases} \end{aligned}$$

Now consider the Lagrangian and dual function of Problem (1.2)

$$\begin{aligned} \mathcal{L}(X, \lambda) &= \langle C, X \rangle - \sum_{i=1}^m \lambda_i (\langle A_i, X \rangle - b_i), \\ g(y) &= \inf_{X \in \mathbb{S}_+^n} \mathcal{L}(X, \lambda) = \sum_{i=1}^m \lambda_i b_i + \inf_{X \in \mathbb{S}_+^n} \left\langle \underbrace{\left(C - \sum_{i=1}^m \lambda_i A_i \right)}_{S}, X \right\rangle = \begin{cases} b^\top \lambda & \text{if } S \succeq 0, \\ -\infty & \text{otherwise.} \end{cases} \end{aligned}$$

One can easily check that the dual problems of Problem (1.1) and Problem (1.2) are both

$$\begin{aligned} &\max_{\lambda \in \mathbb{R}^m} \ b^\top \lambda \\ &\text{s.t. } C - \sum_{i=1}^m \lambda_i A_i \succeq 0 \end{aligned} \tag{1.3}$$

By weak duality, we have $d^* \le p^*_{\text{SDP}} \le p^*_{\text{QCQP}}$. Now assume that strong duality holds, we have that $S \succeq 0$ by dual feasibility and $x^\top S x = 0$ by the dual function, which implies $Sx = 0$.

Theorem 1.2

Let $\bar{x}$ and $\bar{\lambda}$ be a feasible solution of Problem (1.1) and Problem (1.3), respectively. If

$$S_{\bar{\lambda}} := C - \sum_{i=1}^m \bar{\lambda}_i A_i \succeq 0 \quad \text{and} \quad S_{\bar{\lambda}} \bar{x} = 0,$$

then $\bar{x}$ and $\bar{\lambda}$ are a global minimum and global maximum, respectively.

In practice, suppose we have obtained $\bar{x}$ by some solver. We solve the linear system $S_{\bar{\lambda}} \bar{x} = 0$ to obtain a candidate $\bar{\lambda}$, then check the condition $S_{\bar{\lambda}} \succeq 0$.

2 Moment relaxation

Definition 2.1: Polynomials

A polynomial p in $x_1, \dots, x_n$ with coefficients in $\mathbb{R}$ is a finite linear combination of the form $p = \sum_\alpha p_\alpha x^\alpha$, where $\alpha \in \mathbb{N}^n$. The set of all polynomials in $\mathbb{R}$ is denoted by $\mathbb{R}[x]$.

We first review some basic properties. A polynomial in n variables of degree d has C_d^{n+d} coefficients. We denote the standard *monomial basis* in x up to degree d as $[x]_d$. Define the coefficient vector $c_p := (p_\alpha, \dots)^\top$, we can write the polynomial as $p(x) = c_p^\top [x]_d$. For example,

$$h(x) = 1 - x_1^2 - x_2^2 = 1 \cdot x_1^0 x_2^0 - 1 \cdot x_1^2 x_2^0 - 1 \cdot x_1^0 x_2^2 = c_h^\top [x]_2,$$

where

- $\alpha \in \{(0,0),(1,0),(0,1),(2,0),(1,1),(0,2)\} = \{\alpha \in \mathbb{N}^2 \mid |\alpha| \le 2\}$,
- $c_h = (1,0,0,-1,0,-1)^\top$,
- $[x]_2 = (x_1^0x_2^0, x_1^1x_2^0, x_1^0x_2^1, x_1^2x_2^0, x_1^1x_2^1, x_1^0x_2^2)^\top = (1, x_1^1, x_2^1, x_1^2, x_1^1x_2^1, x_2^2)^\top$,
- the dimension of c_h and $[x]_2$ are $C_2^{2+2} = 6$.

Note that $\mathbb{R}[x]$ is a vector space, and the subspace of polynomials up to degree d is denoted as $\mathbb{R}[x]_d$, with dimension $\mathbb{R}^{C_d^{n+d}}$. Now consider a polynomial optimization problem (POP) where the objective and constraints are all polynomials:

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \ & f(x) \\ \text{s.t. } & g_i(x) \ge 0, \ i = 1, \dots, l_g \\ & h_i(x) = 0, \ i = 1, \dots, l_h \end{aligned} \tag{2.1}$$

Define the semialgebraic set that represents the constraints

$$\mathbb{K}_c = \{x \in \mathbb{R}^n \mid g_i(x) \ge 0, \ i = 1, \dots, l_g, \ h_i(x) = 0, \ i = 1, \dots, l_h\}, \tag{2.2}$$

then Problem (2.1) is equivalent to minimizing $f(x)$ over $\mathbb{K}_c$.

2.1 Moment problem

Definition 2.2: Measures

Let Σ be a collection of measurable sets, then the set function $\mu : \Sigma \to \mathbb{R} \cup \{\pm\infty\}$ is called a *measure* if (i) $\mu(\emptyset) = 0$, (ii) $\mu(A) \ge 0$ for all $A \in \Sigma$ (non-negativity), and (iii) $\mu(A \cup B) = \mu(A) + \mu(B)$ if A, B are disjoint sets (countable additivity).

The moment problem is to find measure μ given some moments m_n, where m_n denotes the μ-weighted average of x^n:

$$m_n = \int_{-\infty}^{\infty} x^n \, d\mu.$$

Example 2.3

Given the moments $m_0 = \int_\mathbb{R} 1 \, d\mu = 1$ (sum of weights), $m_1 = \int_\mathbb{R} x \, d\mu = 2$ (average weight of x), and $m_2 = \int_\mathbb{R} x^2 \, d\mu = 5$ (average weight of x^2). Recall that the Dirac measure is defined as

$$\delta_x(A) = \begin{cases} 0, & x \notin A; \\ 1, & x \in A. \end{cases}$$

We can construct $\mu = \frac{1}{2}\delta_1 + \frac{1}{2}\delta_3$. One can verify that $m_0 = \frac{1}{2} + \frac{1}{2} = 1$, $m_1 = \frac{1}{2} + \frac{3}{2} = 2$, and $m_2 = \frac{1}{2} + \frac{3^2}{2} = 5$. Alternatively, one can also check that the measure $\mu = \frac{1}{8}\delta_0 + \frac{3}{4}\delta_2 + \frac{1}{8}\delta_4$ is a solution. That is, solution need not to be unique.

Let $\mathbb{K}$ be a Borel subset of $\mathbb{R}^n$, and the positive cone $\mathscr{M}(\mathbb{K})_+$ be the space of finite Borel measures μ on $\mathbb{K}$. These are just convenient notations where the Borel subset is a canonical way to define *measurable* subsets in $\mathbb{R}^n$. Then the POP (2.1) is equivalent to the moment problem:

$$\begin{aligned} \min_{\mu \in \mathscr{M}(\mathbb{K}_c)_+} & \int_{\mathbb{K}_c} f(x)\ d\mu \\ \text{s.t.} & \int_{\mathbb{K}_c} 1\ d\mu = 1 \end{aligned} \tag{2.3}$$

The constraint forces the measure to be a *probability measure*, i.e., the sum of weights equal to 1. Without this constraint, we can put an arbitrarily large weight to any point x where $f(x) < 0$ and make the objective infinitely small. One optimal measure μ^* is the Dirac measure δ_{x^*} where we put the weight 1 on $f(x^*)$ and 0 for all other points.

2.2 Linear functional representation

Instead of optimizing over the measure μ in Problem (2.3), we opt to optimize with moments instead. Let $y = (y_\alpha, \dots)^\top$ be the moment sequence such that we have the *moment mapping* for all $\alpha \in \mathbb{N}^n$

$$y_\alpha = \int_{\mathbb{K}} x^\alpha\ d\mu.$$

However, this moment sequence is infinite-dimensional and intractable, thereby we set a degree bound $2\kappa \geq d$ such that $y = (y_\alpha, \dots)^\top \in \mathbb{R}^{C_{2\kappa}^{n+2\kappa}}$ for $\alpha \in \{\alpha \in \mathbb{N}^n \mid |\alpha| \leq 2\kappa\}$. Then we have

$$\begin{aligned} \int_{\mathbb{K}} f(x)\ d\mu &= \int_{\mathbb{K}} c_f^\top [x]_{2\kappa}\ d\mu = c_f^\top y, \\ \int_{\mathbb{K}} 1\ d\mu &= \int_{\mathbb{K}} [x]_{2\kappa}^{(1)}\ d\mu = y^{(1)} = 1. \end{aligned}$$

One would have to adjust the dimension of c_f to match with the dimension of y by simply setting the components that do not appear in $[x]_d$ as 0. We continue to use c_f given any degree bound 2κ for notation simplicity. Denote the moment mapping linear functional as

$$L_y : \mathbb{R}[x]_{2\kappa} \to \mathbb{R}\ ; \quad L_y(f(x)) = L_y(c_f^\top [x]_{2\kappa}) = c_f^\top y,$$

we can then reformulate an equivalent problem for Problem (2.3):

$$\begin{aligned} \min_{y \in \mathbb{R}^{C_{2\kappa}^{n+2\kappa}}} \quad & c_f^\top y \\ \text{s.t.} \quad & e_1^\top y = 1 \\ & L_y(f) = y \end{aligned} \tag{2.4}$$

Note that this is a relaxation and the relaxation becomes tighter as $\kappa \to \infty$. Problem (2.4) is nice because the objective and the first constraint are both linear, and the remaining issue is how to state the conditions for the linear functional. We next introduce the *moment matrix* and *localizing matrices* to encode $L_y(f) = y$ as semidefinite constraints.

2.3 Moment and Localizing matrices

The idea of moment matrix is to ensure there is a valid measure μ for the moment mapping on some measurable set $\mathbb{K}$, while localizing matrices are for the measure μ to be valid on the semialgebraic set $\mathbb{K}_c$ defined in (2.2).

Definition 2.4: Moment Matrix

The κ-th order moment matrix of $y \in \mathbb{R}^{C_{2\kappa}^{n+2\kappa}}$ is

$$M_\kappa[y] = L_y([x]_\kappa [x]_\kappa^\top).$$

Each component $M_\kappa[y]_{ij} = L_y(x^\alpha x^\beta) = L_y(x^{\alpha+\beta}) = y_{\alpha+\beta}$, where the i-th and j-th componentn of $[x]_\kappa$ are x^α and x^β, respectively.

For any polynomial $p(x) = c_p^\top [x]_\kappa \in \mathbb{R}[x]_\kappa$, we have

$$c_p^\top M_\kappa[y] c_p = L_y(c_p^\top [x]_\kappa [x]_\kappa^\top c_p) = L_y(p^2) = \int_{\mathbb{K}} p^2 \, d\mu \geq 0.$$

This means that $M_\kappa[y] \succeq 0$ is a necessary condition for μ to be a valid measure on $\mathbb{K}$.

Example 2.5

Let $x \in \mathbb{R}^2$ and $[x]_1 = (1, x_1, x_2)^\top$, the moment matrix is

$$M_1[y] = L_y \begin{pmatrix} 1 & x_1 & x_2 \\ x_1 & x_1^2 & x_1 x_2 \\ x_2 & x_1 x_2 & x_2^2 \end{pmatrix} = L_y \begin{pmatrix} x_1^0 x_2^0 & x_1^1 x_2^0 & x_1^0 x_2^1 \\ x_1^1 x_2^0 & x_1^2 x_2^0 & x_1^1 x_2^1 \\ x_1^0 x_2^1 & x_1^1 x_2^1 & x_1^0 x_2^2 \end{pmatrix} = \begin{pmatrix} y_{00} & y_{10} & y_{01} \\ y_{10} & y_{20} & y_{11} \\ y_{01} & y_{11} & y_{02} \end{pmatrix}.$$

Note that $M_\kappa[y]$ actually constrains monomials in $[x]_{2\kappa}$, which is why the degree bound is set to 2κ in order to simplify notations.

Definition 2.6: Localizing Matrix

Given a polynomial $u \in \mathbb{R}[x]_{2\kappa}$, the κ-th order localizing matrix of $y \in \mathbb{R}^{C_{2\kappa}^{n+2\kappa}}$ is

$$M_{\kappa-d_u}[uy] = L_y(u(x)[x]_\kappa [x]_\kappa^\top),$$

where $d_u = \lceil \deg(u)/2 \rceil$. Each component $M_{\kappa-d_u}[uy]_{ij} = \sum_k (c_u)_k y_{\alpha+\beta+\gamma}$, where c_u is the coefficient vector of u with the k-th component of $[x]_{\deg(u)}$ is γ, the i-th and j-th component of $[x]_\kappa$ are x^α and x^β, respectively.

Consider polynomials $g(x), h(x) \in \mathbb{R}[x]$ such that $g(x) \geq 0$ and $h(x) = 0$. Again, given polynomials $p^g(x) \in \mathbb{R}[x]_{\kappa-d_g}$ and $p^h(x) \in \mathbb{R}[x]_{\kappa-d_h}$, we have

$$c_{p^g}^\top M_{\kappa-d_g}[gy] c_{p^g} = L_y(g(x) \cdot c_{p^g}^\top [x]_{\kappa-d_g} [x]_{\kappa-d_g}^\top c_{p^g}) = L_y(g(p^g)^2) \geq 0,$$
$$c_{p^h}^\top M_{\kappa-d_h}[hy] c_{p^h} = L_y(h(x) \cdot c_{p^h}^\top [x]_{\kappa-d_h} [x]_{\kappa-d_h}^\top c_{p^h}) = L_y(h(p^h)^2) = 0.$$

This means that $M_{\kappa-d_{g_i}}[g_i y] \succeq 0$ for $i = 1, \ldots, l_g$ and $M_{\kappa-d_{h_i}}[h_i y] = 0$ for $i = 1, \ldots, l_h$ are necessary conditions for μ to be a valid measure on the semialgebraic set $\mathbb{K}_c$ defined in (2.2).

Example 2.7

Let $x \in \mathbb{R}^2$, consider the equality constraint $h(x) = 1 - x_1^2 - x_2^2$. Then the second order localizing matrix is

$$\begin{aligned}
M_{2-\lceil 2/2 \rceil}[hy] = M_1[hy] &= L_y(h(x)[x]_1[x]_1^\top) \\
&= L_y\left((1-x_1^2-x_2^2)\begin{pmatrix} 1 & x_1 & x_2 \\ x_1 & x_1^2 & x_1x_2 \\ x_2 & x_1x_2 & x_2^2 \end{pmatrix}\right) \\
&= L_y\begin{pmatrix} 1-x_1^2-x_2^2 & x_1-x_1^3-x_1x_2^2 & x_2-x_1^2x_2-x_2^3 \\ x_1-x_1^3-x_1x_2^2 & x_1^2-x_1^4-x_1^2x_2^2 & x_1x_2-x_1^3x_2-x_1x_2^3 \\ x_2-x_1^2x_2-x_2^3 & x_1x_2-x_1^3x_2-x_1x_2^3 & x_2^2-x_1^2x_2^2-x_2^4 \end{pmatrix} \\
&= \begin{pmatrix} y_{00}-y_{20}-y_{02} & y_{10}-y_{30}-y_{12} & y_{01}-y_{21}-y_{03} \\ y_{10}-y_{30}-y_{12} & y_{20}-y_{40}-y_{22} & y_{11}-y_{31}-y_{13} \\ y_{01}-y_{21}-y_{03} & y_{11}-y_{31}-y_{13} & y_{02}-y_{22}-y_{04} \end{pmatrix}.
\end{aligned}$$

Similarly, $M_{\kappa-d_u}[uy]$ constrains monomials in $[x]_{2\kappa}$. Altogether, we have the κ-th moment relaxation of the POP (2.1):

$$\begin{aligned}
\min_{y \in \mathbb{R}^{C_{2\kappa}^{n+2\kappa}}} \quad & c_f^\top y \\
\text{s.t.} \quad & e_1^\top y = 1 \\
& M_\kappa[y] \succeq 0 \\
& M_{\kappa-d_{g_i}}[g_i y] \succeq 0, \ i = 1, \dots, l_g \\
& M_{\kappa-d_{h_i}}[h_i y] = 0, \ i = 1, \dots, l_h
\end{aligned} \tag{2.5}$$

where $d_{g_i} = \lceil \deg(g_i)/2 \rceil$ and $d_{h_i} = \lceil \deg(h_i)/2 \rceil$.

3 SOS relaxation

In this section, we give another approach to obtain Problem (2.5) by the sum of squares (SOS) relaxation. We first reformulate the POP (2.1) to an equivalent problem:

$$\begin{aligned}
\max_{\gamma \in \mathbb{R}} \ & \gamma \\
\text{s.t.} \ & f(x) - \gamma \geq 0, \ x \in \mathbb{K}_c
\end{aligned} \tag{3.1}$$

The main idea is to certify whether $f(x) - \gamma$ is positive on the semialgebraic set $\mathbb{K}_c$ defined in (2.2) by some *Positivestellensatz*. Before we get into the certificates, we first introduce the SOS polynomials and construct some representation sets.

3.1 Positivestellensatz

Definition 3.1

Let $d = 2s$, a polynomial $p(x) \in \mathbb{R}[x]_d$ is a sum of squares (SOS) polynomial if there exists $q_1, \dots, q_m \in \mathbb{R}[x]_s$ such that

$$p(x) = \sum_{k=1}^{m} q_k^2(x).$$

Let $G = \{g_i\}_{i=1}^{l_g}$ and $H = \{h_i\}_{i=1}^{l_h}$ be the set of inequality and equality constraints, respectively. With the help of SOS polynomials, we now construct a superset of G such that for any polynomial g in the superset, $g(x) \geq 0$ if x is in the feasible set of G. The *quadratic module* of G is defined as follows:

$$\mathbf{qmodule}[G] = \left\{ \sigma_0 + \sum_{i=1}^{l_g} \sigma_i g_i \mid \sigma_i \text{ are SOS} \right\}.$$

Another larger superset of G is the *preorder*, which is defined as follows:

$$\mathbf{preorder}[G] = \left\{ \sigma_0 + \sum_{\{i\}} \sigma_i g_i + \sum_{\{i,j\}} \sigma_{ij} g_i g_j + \cdots \mid \sigma_i, \sigma_{ij}, \ldots \text{ are SOS} \right\}.$$

Similarly, we can also construct the *ideal* of H such that for any polynomial h in the ideal, $h(x) = 0$ if x is in the feasible set of H:

$$\mathbf{ideal}[H] = \left\{ \sum_{i=1}^{l_h} t_i h_i \right\}.$$

Theorem 3.2: Positivestellensatz

Let $\mathbb{K}_c$ be the semialgebraic set defined in (2.2). Then the set $\mathbb{K}_c$ is empty if and only if there exists $g \in \mathbf{preorder}[G]$ and $h \in \mathbf{ideal}[H]$, such that $g + h + 1 = 0$.

Theorem 3.2 ([1, Theorem 3.127]) states that for any infeasible system of equalities and inequalities (i.e., $\mathbb{K}_c = \emptyset$), there exists a simple certificate. We next generalize this result to certify whether a given polynomial is positive on $\mathbb{K}_c$ or not.

Theorem 3.3: Stengle's Positivestellensatz

Let $p(x) \in \mathbb{R}[x]$, then $p(x) > 0$ on $\mathbb{K}_c$ if and only if there exist $g_1, g_2 \in \mathbf{preorder}[G]$ and $h \in \mathbf{ideal}[H]$ such that

$$p = \frac{1 + g_1 + h}{g_2}.$$

Proof. Showing $p(x) > 0$ is equivalent to showing $-p(x) \geq 0$ is infeasible. If we add this new constraint into the semialgebraic set such that $\bar{\mathbb{K}}_c = \{x \in \mathbb{K}_c \mid -p(x) \geq 0\}$, then by Theorem 3.2, we have $\bar{\mathbb{K}}_c$ is empty (infeasible) if and only if

$$\begin{aligned}
0 &= \sigma_0 + \sigma_0'(-p) + \sum_{\{i\}} \sigma_i g_i + \sum_{\{i\}} \sigma_i' g_i(-p) + \sum_{\{i,j\}} \sigma_{ij} g_i g_j + \sum_{\{i,j\}} \sigma_{ij}' g_i g_j(-p) + \cdots + \sum_{i=1}^{l_h} t_i h_i + 1 \\
&= \underbrace{\sigma_0 + \sum_{\{i\}} \sigma_i g_i + \sum_{\{i,j\}} \sigma_{ij} g_i g_j + \cdots}_{g_1 \in \mathbf{preorder}[G]} - p \underbrace{\left(\sigma_0 + \sum_{\{i\}} \sigma_i g_i + \sum_{\{i,j\}} \sigma_{ij} g_i g_j + \cdots \right)}_{g_2 \in \mathbf{preorder}[G]} + \underbrace{\sum_{i=1}^{l_h} t_i h_i}_{h \in \mathbf{ideal}[H]} + 1 \\
&= g_1 - p g_2 + h + 1.
\end{aligned}$$

□

We can add additional assumptions to simplify Theorem 3.3. For example, if we assume $\mathbb{K}_c$ is *compact*, we have the Schmüdgen's Positivestellensatz. We state a more commonly used result by assuming additional Archimedean property.

Lemma 3.4: Archimedean

Let $\mathcal{D} = \mathbf{qmodule}[G] + \mathbf{ideal}[H]$, then $\mathcal{D}$ is *Archimedean* if there exists $N \in \mathbb{N}$ such that the polynomial $N - \sum_{i=1}^{n} x_i^2$ is in $\mathcal{D}$.

Theorem 3.5: Putinar's Positivestellensatz

Let $p(x) \in \mathbb{R}[x]$ and $\mathbb{K}_c$ be compact. Assume that $\mathbf{qmodule}[G]+\mathbf{ideal}[H]$ is Archimedean, then $p(x) > 0$ on $\mathbb{K}_c$ if and only if there exists $g \in \mathbf{qmodule}[G]$ and $h \in \mathbf{ideal}[H]$ such that $p = g + h$, i.e., $p \in \mathbf{qmodule}[G] + \mathbf{ideal}[H]$.

With the additional assumptions, by Theorem 3.5, we can rewrite Problem (3.1) as follows:

$$\begin{aligned} &\max_{\gamma \in \mathbb{R}} \ \gamma \\ &\ \text{s.t. } f(x) - \gamma \in \mathbf{qmodule}[G] + \mathbf{ideal}[H] \end{aligned} \tag{3.2}$$

Note that Problem (3.2) is still intractable, we add a degree bound 2κ such that $\deg(g) \leq 2\kappa$ for all $g \in \mathbf{qmodule}[G]$ and $\deg(h) \leq 2\kappa$ for all $h \in \mathbf{ideal}[H]$, respectively. Therefore, we relaxed Problem (3.2) as

$$\begin{aligned} &\max_{\gamma \in \mathbb{R}} \ \gamma \\ &\ \text{s.t. } f(x) - \gamma \in \mathbf{qmodule}[G]_{2\kappa} + \mathbf{ideal}[H]_{2\kappa} \end{aligned} \tag{3.3}$$

3.2 Conic duality

Problem (3.3) is also known as the κ-th order SOS relaxation of the POP (2.1). It turns out that the SOS relaxation (3.3) is the dual problem of the moment relaxation (2.5). Let $\mathcal{K}$ and $\mathcal{D}$ be the cones in the moment relaxation (2.5) and SOS relaxation (3.3), respectively:

$$\begin{aligned} \mathcal{K} = \{y \in \mathbb{R}^{C_{2\kappa}^{n+2\kappa}} \mid\ & M_\kappa[y] \succeq 0, \\ & M_{\kappa - d_{g_i}}[g_i y] \succeq 0,\ i = 1, \ldots, l_g, \\ & M_{\kappa - d_{h_i}}[h_i y] = 0,\ i = 1, \ldots, l_h\}, \\ \mathcal{D} = \mathbf{qmodule}&[G]_{2\kappa} + \mathbf{ideal}[H]_{2\kappa}. \end{aligned}$$

Recall the definition of dual cones (in $\mathbb{R}^{C_{2\kappa}^{n+2\kappa}}$):

$$\mathcal{K}^* = \{c_p \in \mathbb{R}^{C_{2\kappa}^{n+2\kappa}} \mid c_p^\top y \geq 0,\ \forall y \in \mathcal{K}\}.$$

We claim that c_p is the coefficient vector of the polynomials in $\mathcal{D}$, that is,

$$\mathcal{K}^* = \{c_p \in \mathbb{R}^{C_{2\kappa}^{n+2\kappa}} \mid p(x) = c_p^\top [x]_{2\kappa} \in \mathbf{qmodule}[G]_{2\kappa} + \mathbf{ideal}[H]_{2\kappa}\}.$$

Proof. Since y is constrained by the moment and localizing matrices, we have $c_p^\top y = L_y(p)$ where L_y is the moment mapping. Then we have

$$c_p^\top y = L_y(\sigma_0) + \sum_{i=1}^{l_g} L_y(\sigma_i g_i) + \sum_{i=1}^{l_h} L_y(t_i h_i). \tag{3.4}$$

All three terms in Equation (3.4) are greater or equal to 0. The first term

$$L_y(\sigma_0) = L_y\left(\sum_{k=1}^{m} q_k^2\right) = \sum_{k=1}^{m} L_y([x]_\kappa^\top c_{q_k} c_{q_k}^\top [x]_\kappa) = \sum_{k=1}^{m} c_{q_k}^\top L_y([x]_\kappa [x]_\kappa^\top) c_{q_k} = \sum_{k=1}^{m} c_{q_k}^\top M_\kappa[y] c_{q_k} \geq 0,$$

the second term

$$\sum_{i=1}^{l_g} L_y(\sigma_i g_i) = \sum_{i=1}^{l_g} L_y\left(g_i \sum_{k=1}^{m_i} q_{i,k}^2\right) = \sum_{i=1}^{l_g}\left(\sum_{k=1}^{m_i} L_y(g_i[x]_{\kappa-d_{g_i}}^\top c_{q_{i,k}} c_{q_{i,k}}^\top [x]_{\kappa-d_{g_i}})\right)$$
$$= \sum_{i=1}^{l_g}\left(\sum_{k=1}^{m_i} c_{q_{i,k}}^\top L_y(g_i[x]_{\kappa-d_{g_i}}[x]_{\kappa-d_{g_i}}^\top) c_{q_{i,k}}\right)$$
$$= \sum_{i=1}^{l_g}\left(\sum_{k=1}^{m_i} c_{q_{i,k}}^\top M_{\kappa-d_{g_i}}[g_i y] c_{q_{i,k}}\right) \geq 0,$$

and the last term

$$\sum_{i=1}^{l_h} L_y(t_i h_i) = \sum_{i=1}^{l_h} L_y(h_i c_{t_i}^\top [x]_{\kappa-d_{h_i}}) = \sum_{i=1}^{l_h} c_{t_i}^\top L_y(h_i [x]_{\kappa-d_{h_i}}) = \sum_{i=1}^{l_h} c_{t_i}^\top \sum_k (c_{h_i})_k y_{\alpha+\gamma},$$

where the k-th component of $[x]_{\deg(h_i)}$ is γ, the i-th component of $[x]_{\kappa-d_{h_i}}$ is x^α. Note that the constraint $M_{\kappa-d_{h_i}}[h_i y] = 0$ states that each component

$$M_{\kappa-d_{h_i}}[h_i y]_{ij} = \sum_k (c_{h_i})_k y_{\alpha+\beta+\gamma} = 0,$$

where the j-th component of $[x]_{\kappa-d_{h_i}}$ is x^β. Take β as $(0,\ldots,0)$, we have $\sum_k (c_{h_i})_k y_{\alpha+\gamma} = 0$, and thus the last term equals to 0. Altogether, $c_p^\top y \geq 0$ for any $y \in \mathcal{K}$. $\square$

Now recall the primal-dual conic programs and our moment and SOS relaxations:

$$\text{(P)} \quad \begin{aligned} \min_{x\in\mathbb{R}^n} \ & c^\top x \\ \text{s.t.} \ & Ax = b \\ & x \in \mathcal{K} \end{aligned} \qquad\qquad \text{(D)} \quad \begin{aligned} \max_{y\in\mathbb{R}^m} \ & b^\top y \\ \text{s.t.} \ & c - A^\top y \in \mathcal{K}^* \end{aligned}$$

$$\text{(2.5)} \quad \begin{aligned} \min_{y\in\mathbb{R}^{C_{2\kappa}^{n+2\kappa}}} \ & c_f^\top y \\ \text{s.t.} \ & e_1^\top y = 1 \\ & y \in \mathcal{K} \end{aligned} \qquad\qquad \text{(3.3)} \quad \begin{aligned} \max_{\gamma\in\mathbb{R}} \ & \gamma \\ \text{s.t.} \ & c_f - \gamma e_1 \in \mathcal{K}^* \end{aligned}$$

where the constraint $c_f - \gamma e_1 \in \mathcal{K}^*$ is equivalent to $f(x) - \gamma \in \mathbf{qmodule}[G]_{2\kappa} + \mathbf{ideal}[H]_{2\kappa}$ and $c_f - \gamma e_1$ is the coefficient vector of $f(x) - \gamma$.

4 Lasserre hierarchy

Let ρ be the optimal value of the POP (2.1), p_κ^* and d_κ^* be the optimal value of the κ-th moment relaxation (2.5) and κ-th SOS relaxation (3.3), respectively. By weak duality, we have

$$d_\kappa^* \leq p_\kappa^* \leq \rho$$

for all κ. Moreover, both $\{p_\kappa^*\}_{\kappa=1}^\infty$ and $\{d_\kappa^*\}_{\kappa=1}^\infty$ are monotone increasing sequences. As $\kappa \to \infty$, the moment relaxation recovers the moment problem (2.3), and Putinar's Positivestellensatz holds for the SOS relaxation. In other words, suppose that $\mathbb{K}_c$ is compact and the Archimedean property holds, then

$$\lim_{\kappa\to\infty} d_\kappa^* = \lim_{\kappa\to\infty} p_\kappa^* = \rho.$$

The idea of Lasserre's hierarchy is quite intuitive, we solve a sequence of κ-th moment relaxations and increase κ until some stopping criteria. We next introduce the *flat extension*, a sufficient condition for verifying whether a moment relaxation is tight or not, as well as a method for extracting solutions x^* given y^*.

4.1 Flat extension

The intuition is that suppose we have solved a moment relaxation problem and obtained $M_\kappa[y]$ with $\mathrm{rank}(M_\kappa[y]) = r$. If we extend the degree by 1 such that $\mathrm{rank}(M_{\kappa+1}[y]) = r$, this means that the additional moments are already determined by the old ones. For example, let $\mu = \delta_0 + \delta_1$ be a measure on $\mathbb{R}$, we have the moment sequence

$$y = (y_0, y_1, y_2, \dots) = (1 \cdot 0^0 + 1 \cdot 1^0, 1 \cdot 0^1 + 1 \cdot 1^1, 1 \cdot 0^2 + 1 \cdot 1^2, \dots) = (2, 1, 1, \dots).$$

The first and second order moment matrices are

$$M_1[y] = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad M_2[y] = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix},$$

and $\mathrm{rank}(M_1[y]) = \mathrm{rank}(M_2[y]) = 2$. Such $M_{\kappa+1}[y]$ is called the *flat extension* of $M_\kappa[y]$. For our moment relaxation problem, we need a more general condition to account for the constraints. Let

$$\begin{aligned} v &= \max\{1, \lceil \deg(g_i)/2 \rceil, i = 1, \dots, l_g, \ \lceil \deg(h_i)/2 \rceil, i = 1, \dots, l_h\}, \\ \kappa_0 &= \max\{v, \lceil \deg(f)/2 \rceil\}. \end{aligned} \tag{4.1}$$

We solve the κ-th order moment relaxation where $\kappa \geq \kappa_0$ ensures we have enough moments to represent the objective and constraints.

Theorem 4.1

Let v and κ_0 be as defined in (4.1), $\kappa \geq \kappa_0$, ρ and p_κ^* be the optimal value of the POP and the κ-th moment relaxation, respectively. If the moment relaxation has an optimal solution y^* such that

$$\mathrm{rank}(M_s[y^*]) = \mathrm{rank}(M_{s-v}[y^*])$$

for some $s \in [\kappa_0, \kappa]$, then $p_\kappa^* = \rho$ (i.e., the relaxation is tight).

If Theorem 4.1 holds, then we have that $c_f^\top y^* = c_f^\top [x]_d = f(x^*)$. Yet, we still do not know what the optimal solution to the POP x^* is. We next introduce the method of extracting solutions [4].

4.2 Extraction of solutions

In this section, we assume that Theorem 4.1 holds with $\mathrm{rank}(M_s[y^*]) = r$.

4.2.1 Unique solution

We first consider the case where $r = 1$. If the global minimum x^* of the POP is unique, then the optimal measure of the moment problem $\mu^* = \delta_{x^*}$ is unique. Furthermore, y^* is also unique and

$$M_s[y^*] = L_y([x]_s [x]_s^\top) = [x^*]_s [x^*]_s^\top,$$

where $[x^*]_s$ is the monomial basis $[x]_s$ evaluated at x^*. That is, if $\mathrm{rank}(M_s[y^*]) = 1$, then we can extract the solution by eigendecomposition where $[x^*]_s$ is the eigenvector corresponding to the single positive eigenvalue, and x^* is obtained by reading the degree one monomials of $[x^*]_s$.

4.2.2 Multiple solutions

Now consider the case where $r > 1$. If there are r global minimums $x^*(1), \dots, x^*(r)$ of the POP, then the measures $\delta_{x^*(1)}, \dots, \delta_{x^*(r)}$ are all optimal measures of the moment problem. In addition, every convex combination $\sum_j \theta_j \delta_{x^*(j)}$ where $\sum_j \theta_j = 1$ and $\theta_j \geq 0$ is also an optimal measure. That is, the moment problem admits an r-atomic measure [3, Theorem 1.6],

$$\begin{aligned} M_s[y^*] &= \sum_{j=1}^{r} \theta_j [x^*(j)]_s [x^*(j)]_s^\top \\ &= \underbrace{\begin{pmatrix} [x^*(1)]_s & \cdots & [x^*(r)]_s \end{pmatrix}}_{V^* \in \mathbb{R}^{C_{2s}^{n+2s} \times r}} \underbrace{\begin{pmatrix} \theta_1 & & \\ & \ddots & \\ & & \theta_r \end{pmatrix}}_{D \in \mathbb{R}^{r \times r}} \begin{pmatrix} [x^*(1)]_s^\top \\ \vdots \\ [x^*(r)]_s^\top \end{pmatrix} = V^* D (V^*)^\top. \end{aligned}$$

The first step is to obtain V^* given $M_s[y^*]$. We use eigendecomposition to compute the Cholesky factor $V \in \mathbb{R}^{C_{2s}^{n+2s} \times r}$ such that $M_s[y^*] = VV^\top$. Note that the columns of V and V^* span the same column space, we next compute the reduced column echelon form of V:

$$U = \begin{pmatrix} 1 & & & \\ * & & & \\ \vdots & & & \\ 0 & 1 & & \\ * & * & & \\ \vdots & \vdots & \ddots & \\ 0 & 0 & \cdots & 1 \\ * & * & \cdots & * \\ \vdots & \vdots & & \vdots \\ * & * & \cdots & * \end{pmatrix} \in \mathbb{R}^{C_{2s}^{n+2s} \times r}$$

Since the columns of U still span the same column space as V^*, we have that for any global minimum x^*, $[x^*]_s = Uw$ for some coefficient vector $w \in \mathbb{R}^r$. By the rows of the pivot points, we have $[x^*]_s^{(k)} = w^{(k)}$ for $k = 1, \ldots, r$. Thus if we construct the coefficient vector by assigning each component as the monomial that corresponds to the index of pivot points, we have

$$[x^*(j)]_s = U w_{x^*(j)} \tag{4.2}$$

for all $j = 1, \ldots, r$. In other words, if we define $w = (x_{\alpha_1}, \ldots, x_{\alpha_r})^\top$ where α_k are the monomials that correspond to the index of pivot points, then finding $[x^*(j)]_s$ is equivalent to solving the polynomial system of equations

$$[x]_s = Uw \tag{4.3}$$

We next introduce the method of solving Equation (4.3) presented in GlotiPoly [2, 4, 6]. For each degree one monomials, we construct a multiplication matrix N_i such that $N_i w = x_i w$ for $i = 1, \ldots, n$.

TBD. Code available at https://github.com/doggydoggy0101/blog/tree/main/pop.

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